

Suraj Kumar Sahu

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Computational Biophysics | Mechanobiology | Multicellular Dynamics

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RESEARCH PROFILE

Computational biophysicist developing agent-based, particle-based, and active-gel models that connect cell-scale forces and adhesion dynamics to multicellular organization, extracellular-matrix remodeling, and vascular-network function. My research integrates mechanobiology, network analysis, and high-performance simulation to study how collective cell behaviors shape tissue-scale structure and transport.

EDUCATION

- Jan 2021 – Present **PhD Candidate in Physics**, University of California Merced, California, USA.
Advisor: **Prof. Ajay Gopinathan**
- Aug 2017 – May 2019 **Master of Science in Physics**, National Institute of Technology Rourkela, Odisha, India.
- Jul 2014 – Jun 2017 **Bachelor of Science (Honors in Physics)**, D.R. Nayapalli College, Utkal University, Odisha, India.

PUBLICATIONS

Peer-Reviewed Journals

- 2021 **Sahu, S.** and M. Biswas. “Modeling protein association from homogeneous to mixed environments: A reaction-diffusion dynamics approach.” *Journal of Molecular Graphics and Modeling* 107, 107936.

RESEARCH EXPERIENCE

Positions

- Jan 2021 – Present **Graduate Research Assistant**, Gopinathan Group, Department of Physics, University of California Merced.
- Aug 2017 – May 2019 **Graduate Student Researcher**, Computational Biophysics Group, National Institute of Technology Rourkela.
Developed coarse-grained reaction-diffusion simulations in ReaDDy to study macromolecular crowding effects on protein association; showed that crowder size had a larger effect than shape on product stabilization and characterized additive and non-additive effects in mixed-crowder systems, resulting in a peer-reviewed publication.

Selected Doctoral Projects

- Oct 2024 – Present **Compaction of Extracellular Matrix by Multicellular Networks**, in collaboration with **Das-biswas Lab**.
Developed a computational model of collagen-matrix compaction driven by contractile fibroblast networks to study how cell-generated forces reorganize extracellular-matrix structure. *Manuscript in preparation.*
- Aug 2024 – Present **Cortical Tension and Cadherin Adhesion: An Active-Gel Framework**
Developed an active-gel framework for cell-cell contact formation that couples actomyosin-cortex mechanics, cadherin clustering, and external loading to analyze junction formation, maturation, and rupture. *Manuscript in preparation.*

Jan 2024 – Present	Mechanics of Cell Junction Formation, Stability, and Network Remodeling Developed a protrusion-driven two-cell model to test how cadherin-bond kinetics and actomyosin contractility regulate junction stability; extended the framework to multicellular networks to connect junction mechanics with vascular topology. <i>Manuscript in preparation.</i>
Jan 2021 – Present	Agent-Based Modeling of Vasculogenesis , in collaboration with Sindi Lab and Kara E. McCloskey Lab . Developed an agent-based model of cell migration, vascular-network formation, and remodeling; characterized the topology and transport function of the resulting networks. <i>Manuscript in preparation.</i>

TECHNICAL EXPERTISE

Modeling & HPC	Agent-based and particle-based modeling, active-gel mechanics, reaction-diffusion systems, network dynamics, numerical simulation, HPC workload management and batch-job scheduling with Slurm
Programming	Python, Julia
Research Software	NumPy, SciPy, NetworkX, ReaDDy, Git, GitHub, LaTeX, VS Code
Data & Imaging	Data analysis, scientific visualization, image analysis
AI-Enabled Research	Agentic scientific workflow design, AI-assisted computational reproducibility audits, context engineering, Model Context Protocol (MCP) tool development (Cellpose-MCP), AI-enabled image analysis
LLM Research Tools	Claude, Codex

SELECTED ORAL PRESENTATIONS

Feb 2026	Sahu S. , Dasbiswas K., Gopinathan A. “Collective Cell Motility of Fibroblasts Driven by Contractile Multicellular Network Formation during Compaction of Collagen Matrix.” <i>BPS 2026 Annual Meeting</i> , San Francisco, CA.
Mar 2025	Sahu S. , Gopinathan A. “Stability of Cell-Cell Junctions: Balancing Cortical Tension and Cadherin Aggregation at Cell Interface during Cell-Cell Separation.” <i>APS March Meeting 2025</i> , Los Angeles, CA.
Dec 2024	Sahu S. , Gopinathan A. “Balancing Cortical Tension and Adhesive Force for Stable Cell Junctions.” <i>Cell Bio 2024, ASCB EMBO</i> , San Diego, CA.
Mar 2024	Sahu S. , Gopinathan A. “Modeling the Mechanics of Cell-Cell Junction Formation and Dynamics in Vascular Networks.” <i>APS March Meeting 2024</i> , Minneapolis, MN.
Mar 2023	Sahu S. , Gopinathan A., Sindi S., McCloskey K., Kuhn M., Zamora J. “Particle-Based Simulation of the Assembly and Mechanical Remodeling of Vascular Network.” <i>APS March Meeting 2023</i> , Las Vegas, NV.
Mar 2022	Sahu S. , Kuhn M., Zamora J., Gopinathan A., McCloskey K., Sindi S. “Agent-Based Simulation of Vasculogenesis.” <i>APS March Meeting 2022</i> , Chicago, IL.

RESEARCH INSTRUCTION AND PROFESSIONAL DEVELOPMENT

Spring 2025	Organizer and Instructor, Workshop on AI Tools for Research and Data Analysis , University of California Merced Taught context engineering, agentic scientific workflows, MCP tool integration, and research automation for data analysis, image analysis, and literature organization.
Summer 2024	Project Leader and Instructor , Center for Engineering and Mechanobiology Boot Camp, University of Pennsylvania Led a research project and instructed graduate trainees during the annual mechanobiology boot camp.

MENTORING AND TEACHING

- Jun – Aug 2026 **Graduate Student Research Mentor, Interdisciplinary Biological Engineering and Science Training (I-BEST) NSF REU**, University of California Merced
Mentored **Akshay Langhani**, a first-year undergraduate at Illinois State University, on “*Quantifying Biological Functionality of In Silico Vascular Networks.*”
- Jan 2021 – Present **Graduate Teaching Assistant**, Department of Physics, University of California Merced
Led experimental laboratories and discussion sections for PHYS 115, PHYS 009, PHYS 009L, PHYS 018, and PHYS 008L across multiple terms.
- Aug 2024 – Aug 2025 **GradExcel Peer Mentor**, University of California Merced Graduate Division
Mentored incoming graduate students on academic success, personal well-being, and professional development; received the 2024 GradExcel Peer Mentor Award.

FELLOWSHIPS AND AWARDS

- 2025 Center for Engineering Mechanobiology (CEMB) Summer Research Fellowship; Center for Cellular and Biomolecular Machines (CCBM) Travel Award; Physics Graduate Group Travel Fellowship
- 2024 GradExcel Peer Mentor Award; Physics Graduate Group Travel Fellowship
- 2023 CCBM Outreach Fellowship; CCBM Travel Fellowship; Physics Graduate Group Travel Fellowship
- 2022 Bobcat Summer STEM Academy Fellowship; Physics Graduate Group Travel Fellowship

LEADERSHIP AND SERVICE

- Aug 2023 – Aug 2025 **Trainee Leadership Council**, Center for Engineering Mechanobiology (CEMB)
Planned tutorials, research presentations, and professional-development workshops for graduate trainees across multiple institutions.
- Jun 2021 – Jun 2023 **President, Graduate Biophysics Club**, University of California Merced
Led a biophysics journal club and organized science-outreach, professional-development, and networking events for graduate students.

SELECTED SCIENCE OUTREACH

- Sept 2024 **Panelist**, Bahunjan Scholars Network Graduate Application Series
Advised prospective graduate students on application strategy and preparation.
- Jan 2024 **Presenter and Organizer**, Digital Nalanda, Wardha, India
Co-led a Foldscope workshop on exploring microorganisms and the living world.
- Jul 2023 **Organizer**, CCBM Science of Coronavirus Outreach
Organized a school outreach activity using CellPaint to explain coronavirus structure and biology.

REFERENCES

Prof. Ajay Gopinathan
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